

Empirical Calibration of the Bivariate Kou Model: Companion Note

Abstract

This note accompanies *Optimal Sizing of Leveraged Crypto Long–Short Positions under Kou Jump-Diffusion via Killed Moments*. It records the penalized empirical characteristic-function estimator, frequency grid, initialization, and identification considerations underlying the empirical illustration in the main paper.

Model notation

We retain the notation of the main paper. For assets $i = 1, 2$, the free parameter vector is

$$\theta = ((\mu_i, \sigma_i, \lambda_i, p_i, \eta_{i,+}, \eta_{i,-})_{i=1,2}, \rho).$$

Here μ_i is the expected-price growth exponent, $\sigma_i > 0$ is diffusion volatility, $\lambda_i > 0$ is jump intensity, $p_i \in (0, 1)$ is the upward-jump probability, $\eta_{i,+} \in (0, 1)$ and $\eta_{i,-} > 0$ are the mean upward and downward jump sizes, and $|\rho| < 1$ is Brownian correlation. The jump characteristic function and price-jump compensator are

$$\varphi_{J_i}(u) = \frac{p_i}{1 - iu\eta_{i,+}} + \frac{1 - p_i}{1 + iu\eta_{i,-}}, \quad \chi_i = \frac{p_i}{1 - \eta_{i,+}} + \frac{1 - p_i}{1 + \eta_{i,-}} - 1,$$

and the derived log-price drift is $\mu_i^X = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i\chi_i$. Thus

$$\Psi_\theta(u, v) = i(\mu_1^X u + \mu_2^X v) - \frac{1}{2}(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv) + \lambda_1(\varphi_{J_1}(u) - 1) + \lambda_2(\varphi_{J_2}(v) - 1).$$

The liquidation-driving log-price spread is $Z = X_1 - X_2$. If moments through order K are required, the admissibility restriction is $K\eta_{i,+} < 1$ for $i = 1, 2$.

1. Penalized ECF estimator

The closed-form Lévy–Khinchine exponent $\Psi_\theta(u, v)$ gives the model-implied characteristic function

$$\phi_\theta(u, v; \Delta) = \exp(\Delta \Psi_\theta(u, v)), \tag{1}$$

where Δ is the observation interval. Given N observed bivariate log-return pairs $(r_{1,n}, r_{2,n})_{n=1}^N$, the empirical characteristic function is

$$\hat{\phi}_N(u, v) = \frac{1}{N} \sum_{n=1}^N \exp(i(ur_{1,n} + vr_{2,n})). \tag{2}$$

The ECF component of the estimator is the normalized weighted squared distance between (2) and (1) over a discrete frequency grid $\mathcal{G}$:

$$Q_N^{\text{ECF}}(\theta) = \frac{\sum_{(u,v) \in \mathcal{G}} w(u,v) |\hat{\phi}_N(u,v) - \phi_\theta(u,v;\Delta)|^2}{\sum_{(u,v) \in \mathcal{G}} w(u,v)}. \quad (3)$$

Because finite high-frequency samples leave a pronounced intensity–jump-size ridge, the reported calibration uses two transparent soft anchoring terms:

$$\begin{aligned} \hat{\theta} &= \arg \min_{\theta \in \Theta} \mathcal{Q}_N(\theta), \\ \mathcal{Q}_N(\theta) &= Q_N^{\text{ECF}}(\theta) + \omega_\lambda \sum_{i=1}^2 (\log \lambda_i - \log \tilde{\lambda}_i)^2 \\ &\quad + \omega_\eta \sum_{i=1}^2 \sum_{s \in \{+, -\}} (\log \eta_{i,s} - \log \tilde{\eta}_{i,s})^2. \end{aligned} \quad (4)$$

Here $\tilde{\lambda}_i$ is the variance/fourth-cumulant initializer and $\tilde{\eta}_{i,s}$ is the peak-over-threshold (POT) mean-excess initializer described below. The application uses $(\omega_\lambda, \omega_\eta) = (0.05, 0.02)$. These are soft penalties, not equality restrictions; setting both weights to zero recovers pure ECF matching. This is the empirical-characteristic-function approach of Singleton (2001), which is particularly natural for affine and Lévy-based asset-pricing models whose characteristic functions are known in closed form. Feuerverger and McDunnough (1981) establish that ECF estimators over a rich grid can attain maximum-likelihood efficiency. The procedure avoids evaluating the bivariate transition density, which would require a two-dimensional numerical Fourier inversion inside every optimizer call. For background on DEJD likelihood estimation in the univariate case see Ramezani and Zeng (2007).

2. Frequency grid

The parameter vector θ has the 13 free components stated above. Raw return magnitudes depend on the sampling interval, so fixed raw Fourier frequencies are poorly portable. Let s_1 , s_2 , and s_Z be IQR-based robust scales, $s = \max\{(q_{0.75} - q_{0.25})/1.349, 10^{-8}\}$, for r_1 , r_2 , and $r_1 - r_2$. Dimensionless base frequencies are mapped to raw frequencies by division by the corresponding scale. The grid contains the following groups, with every point accompanied by its conjugate $(-u, -v)$:

- marginal points $(a/s_1, 0)$ and $(0, a/s_2)$ for $a \in \mathcal{A}_{\text{marg}}$;
- same-sign and mixed-sign joint points $(a/s_1, b/s_2)$ and $(a/s_1, -b/s_2)$ for $a, b \in \mathcal{A}_{\text{joint}}$, which are informative about ρ ;
- spread points $(a/s_Z, -a/s_Z)$ for $a \in \mathcal{A}_Z$, which directly match the log-ratio process governing liquidation; and
- high-base marginal and spread points for $a \in \mathcal{A}_{\text{tail}}$, included to retain information about jump-tail shape.

The dimensionless base nodes are

$$\begin{aligned}\mathcal{A}_{\text{marg}} &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5, 7.5, 10\}, \\ \mathcal{A}_{\text{joint}} &= \{0.25, 0.5, 1, 2, 3, 5\}, \\ \mathcal{A}_Z &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5\}, \\ \mathcal{A}_{\text{tail}} &= \{8, 12, 16, 20, 25\}.\end{aligned}$$

Points whose absolute raw coordinate exceeds 5,000 are discarded to avoid numerical aliasing. Including conjugate pairs, the unfiltered construction has 236 frequency evaluations; this cap leaves 234 for the application sample. Weights depend on dimensionless rather than raw frequency:

$$w(a, b) = \exp\{-0.02(a^2 + b^2)\}.$$

The ordinary spread group receives an additional factor of two. Thus the criterion emphasizes the barrier-driving ratio without making the choice of observation interval determine the effective frequency weights.

3. Reparameterisation and initialisation

Unconstrained reparameterisation. The parameter constraints stated above are enforced automatically via the change of variables

$$\sigma_i = e^{a_i}, \quad \lambda_i = e^{b_i}, \quad p_i = \text{sigmoid}(c_i), \quad \eta_{i,+} = \eta_{i,+}^{\max} \text{sigmoid}(d_i), \quad \eta_{i,-} = e^{e_i}, \quad \rho = \tanh(f), \quad (5)$$

where $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$. The unconstrained vector $\tau = (\mu_1, a_1, b_1, c_1, d_1, e_1, \mu_2, \dots, f)$ is optimized with box bounds in this transformed space. For both assets the code uses a scaled sigmoid with

$$\eta_{i,+}^{\max} \leq (1 - \varepsilon)/K, \quad i = 1, 2,$$

so that $K\eta_{i,+} < 1$ is enforced for $i = 1, 2$, ensuring finiteness of the exponential moments used through order K .

Threshold initialiser. The characteristic-function objective is non-linear and non-convex, so the optimisation is initialised by a threshold-based decomposition following the logic of [Mancini \(2009\)](#) for discretely observed jump-diffusions. Let $m_i = \text{median}(r_{i,n})$ and let s_i be the IQR robust scale. For a frequency-aware threshold constant $c \in [3, 5]$, define candidate jump observations

$$\mathcal{J}_i = \{n : |r_{i,n} - m_i| > cs_i\}, \quad \mathcal{C}_i = \{1, \dots, N\} \setminus \mathcal{J}_i.$$

Initial volatility and correlation values use the continuous subset $\mathcal{C}_i$, while the jump-size means are anchored by peak-over-threshold excess means and the intensities are corrected by a variance/fourth-cumulant estimate. Schematically:

$$\begin{aligned}p_i^{(0)} &= \frac{|\mathcal{J}_{i,+}|}{|\mathcal{J}_i|} \quad \text{clamped to the admissible probability range,} \\ \eta_{i,+}^{(0)}, \eta_{i,-}^{(0)} &= \text{POT mean-excess estimates,} \\ \lambda_i^{(0)} &= \text{two-moment estimate from variance and fourth cumulant,} \\ \sigma_i^{(0)} &= \text{sd}(r_{i,n} : n \in \mathcal{C}_i) / \sqrt{\Delta}, \quad \rho^{(0)} = \text{corr}(r_1, r_2)_{n \in \mathcal{C}_1 \cap \mathcal{C}_2}.\end{aligned}$$

The initializer first estimates the annualized log-process drift with a damped mean-jump correction,

$$\mu_i^{X,(0)} = \bar{r}_i/\Delta - 0.25 \lambda_i^{(0)} \left[p_i^{(0)} \eta_{i,+}^{(0)} - (1 - p_i^{(0)}) \eta_{i,-}^{(0)} \right],$$

then stores the result in the paper’s price-growth convention,

$$\mu_i^{(0)} = \text{clip}_{[\mu_{\min}, \mu_{\max}]} \left(\mu_i^{X,(0)} + \frac{1}{2} (\sigma_i^{(0)})^2 + \lambda_i^{(0)} \chi_i^{(0)} \right).$$

The threshold filter is not the final estimator. It is a warm start that places the optimizer in a sensible region of parameter space; see the remark in [Mancini \(2009\)](#).

Multi-start cloud. Because $\mathcal{Q}_N(\tau)$ may have local minima, the optimizer is run from a structured cloud. It contains the unperturbed initializer and six deterministic intensity–size ridge starts with

$$\lambda_i \mapsto c\lambda_i, \quad \eta_{i,\pm} \mapsto \eta_{i,\pm}/\sqrt{c}, \quad c \in \{0.1, 0.25, 0.5, 2, 4, 8\}.$$

Remaining starts cycle through four high-jump/high-diffusion scenarios and add bounded random perturbations to scale, probability, correlation, and drift parameters. The application calibration uses 30 starts (the library default is 20) and returns the bounded L-BFGS-B minimizer with the lowest penalized objective.

4. Identifiability and model-comparison protocol

The penalized criterion $\mathcal{Q}_N(\theta)$ is non-convex. Jump-diffusion models possess approximate ridges along directions that preserve the compound-Poisson variance contribution $\lambda_i \mathbb{E}[J_i^2] = 2\lambda_i(p_i \eta_{i,+}^2 + (1 - p_i) \eta_{i,-}^2)$ ([Ramezani and Zeng, 2007](#)). The diffusion volatilities σ_i and the Brownian correlation ρ are generally better identified from short samples than jump-size and intensity parameters, which carry larger finite-sample uncertainty unless the observation window contains many jump events. Because λ_i is annualized, a one-year window contains λ_i jumps in expectation, irrespective of whether prices are sampled hourly or daily. Finer sampling can make individual jumps easier to separate from aggregated returns, but it does not multiply the underlying annual Poisson count.

For descriptive fit comparisons on a common grid, compare the unpenalized component $\mathcal{Q}_N^{\text{ECF}}(\hat{\theta})$ across nested models (GBM, Merton, bivariate Kou), rather than the penalized values $\mathcal{Q}_N(\hat{\theta})$ whose anchors differ. Such comparisons should be accompanied by liquidation-relevant diagnostics: empirical versus model Z-tail quantiles, empirical versus model barrier-crossing frequencies, and simulated versus observed drawdowns of Z. These are more directly informative for the sizing application than standard likelihood-based criteria. The present empirical illustration does not carry out a formal GBM–Merton–Kou selection exercise. Its point calibration and sizing results remain conditional on the selected Kou model and stated empirical choices; they are not evidence of model dominance or parameter-robust recommendations.

The repository’s calibration package contains the implementation. The synthetic calibration job shows parameter recovery and produces [Figure 1](#).

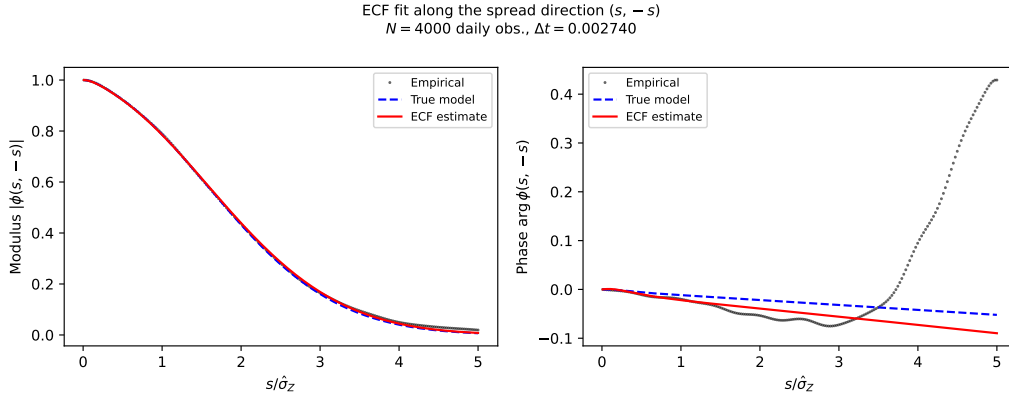


Figure 1: ECF fit along the spread direction $(s, -s)$ for $N = 4,000$ synthetic daily log-return observations. Left: modulus $|\phi(s, -s)|$; right: phase $\arg \phi(s, -s)$. Dots: empirical CF $\hat{\phi}_N$; dashed: true model CF; solid red: ECF estimate. The close agreement in modulus and phase documents the fit along the liquidation-relevant spread direction; this plot alone does not assess every component of the full-grid criterion. Script `jobs/calibrate_kou.py`.

References

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